Mitchell Tillman, PhD

Curriculum Vitae

Ph.D. in Biomedical Engineering from Stevens Institute of Technology in Hoboken, NJ. My work with Dr. Antonia Zaferiou in the Musculoskeletal Control and Dynamics Lab focused on understanding the biomechanics of balance and momentum control during 90-degree turns while walking in healthy younger and older adults. **I am seeking a post-doctoral researcher position** to learn new skills such as computational modeling, image analysis, etc. to support my goal of studying motor control as a tenured professor.

Education

2024 **Ph.D. in Biomed. Eng.**, *Stevens Institute of Technology*, Oct 2024 GPA 4.00/4.0

Thesis: Momentum and Balance Control in Healthy Young and Older Adults During 90-Degree Turns While Walking

2019 **M.S. in Biomechanics**, *Purdue University*, May 2019 GPA 3.59/4.0

Thesis: Effects of Past and Future Motor Events on Present Motor Stability, and Relationships with Motor and Cognitive Flexibility

2018 **B.S. in Movement and Sports Sciences**, *Purdue University*, May 2018GPA 3.13/4.0

Research Positions

2024 **Postdoctoral Research Fellow**, *Shirley Ryan AbilityLab*, *Center for Bionic Medicine*, 2024-Present

- Worked with Dr. Arun Jayaraman studying the effect of spinal cord stimulation on stroke patients

2019 **Research Assistant**, *Stevens Institute of Technology*, *MSKCD Lab*, 2019-2024

- Worked with Dr. Antonia Zaferiou studying the biomechanics of balance and momentum control during turns

- Used MATLAB, MaxMSP, and Optitrack Motive to develop a real-time auditory biofeedback system to assist balance

- Developed open-source MATLAB and Python software to facilitate reproduceable and shareable data analysis workflows.

2016 **Research Assistant**, *Purdue University*, *Biomechanics Lab*, 2016-2018

- Worked with Dr. Satyajit Ambike studying motor control in the Biomechanics Lab in the Health and Kinesiology department

- Helped develop and conduct experiments examining manual dexterity.

- Proficient in biomechanical data acquisition using force transducers and data analysis focusing on movement variability.

Teaching

2021 **Team Lead**, *NJ Governors' STEM Scholars Program*, 2021-2022

- Mentored 10 high school students in a research project I devised to develop a real-time auditory biofeedback system using open source software (DeepLabCut Live, PureData, Python)

- Instructed students on the use of the softwares, organizing group work, scientific writing, mentored students, and ensured delivery of a paper and oral presentation

2018 **Graduate Teaching Assistant**, *Purdue University*, *Health & Kinesiology*, 2018-2019

- Taught two lab sections of upper level undergraduate Exercise Physiology course

- Responsible for leading lab sections and grading lab assignments

Peer-Reviewed Publications

2026 **Tillman M**, Veit N, Kishta A, McKenzie K, Aalla S, Hunt J, Khazanchi R, Petrie TG, Yang C, Lieber R, Roth E, Jayaraman A. "Acute Effects of Transcutaneous Spinal Cord Stimulation (tSCS) on Gait Symmetry in People with Chronic Stroke.", *Neuromodulation: Technology at the Neural Interface*, 2026, DOI: 10.1016/j.neurom.2026.09.018

Veit NC, Aalla S, Kishta A, McKenzie KA, Yang C, **Tillman M**, Roth EJ, Lieber RL, Jayaraman A. "The effects of electrode configuration and location on motor thresholds, tolerance, and muscle selectivity of transcutaneous spinal cord stimulation in stroke.", *Journal of Neurophysiology*, 2026, DOI: 10.1152/jn.00500.2025

Kishta A, Khazanchi R, Veit N, **Tillman M**, McKenzie K, Aalla S, Glenn-Hunt J, Yang C, Privitera L, Yu XS, Chundi J, Lanotte F, Lieber RL, Roth E, Jayaraman A. "Multimodal Machine Learning to Personalize Transcutaneous Spinal Cord Stimulation for Stroke Rehabilitation.", *Digital Biomarkers*, 2026, DOI: 10.1159/000552236

2025 Hirsch ZM, **Tillman M**, Liu J, Molino J, Zaferiou AM. "Older Adult Frontal Plane Balance During 90 Degree Turns while Walking.", *Scientific Reports*, 2025, DOI: 10.1038/s41598-025-22800-x

2024 **Tillman M**, Liu J, Hirsch Z, Molino J, Zaferiou A. "Healthy Older Adults Generate Transverse-Plane Momenta Required for 90° Turns While Walking during the Same Phases of Gait as Used in Straight-Line Gait.", *Journal of NeuroEngineering and Rehabilitation*, 2024

2023 **Tillman M**, Molino J, Zaferiou A. "Gait-Phase Specific Transverse-Plane Momenta Generation during Pre-Planned and Late-Cued 90 Degree Turns While Walking.", *Scientific Reports*, 2023

2022 **Tillman M**, Molino J, Zaferiou A. "Frontal Plane Balance during Pre-Planned and Late-Cued 90 Degree Turns While Walking.", *Journal of Biomechanics*, 2022

2020 **Tillman M**, Ambike S. "The Influence of Recent Actions and Anticipated Actions on the Stability of Finger Forces during a Tracking Task.", *Motor Control*, 2020

2018 **Tillman M**, Ambike S. "Cue-Induced Changes in the Stability of Finger Force-Production Tasks Revealed by the Uncontrolled Manifold Analysis.", *Journal of Neurophysiology*, 2018

Tillman M, Ambike S. "Expectation of Movement Generates Contrasting Changes in Multifinger Synergies in Young and Older Adults.", *Experimental Brain Research*, 2018

Peer-Reviewed Conference Papers and Presentations

2020 **Tillman M**, Dahl L, Knowlton C, Zaferiou A. Real-Time Optical Motion Capture Balance Sonification System. ACM International Conference Proceeding Series, *Virtual*, Jan 2020, *Poster presentation*

Peer-Reviewed Conference Abstracts and Presentations

2026 **Tillman M**, Jayaraman A. Transcutaneous Spinal Cord Stimulation-Augmented Treadmill Gait Training in Chronic Stroke: Single-Session and Preliminary Longitudinal Effects on Step Length. International Conference of NeuroRehabilitation, *Seoul, South Korea*, Sep 2026, *Podium presentation by* **Tillman M**

2025 Khazanchi R, Kishta A, **Tillman M**, Veit N, McKenzie K, Aalla S, Chundi J, Hunt J, Roth EJ, Jayaraman A. Optimization Of Transcutaneous Spinal Cord Stimulation For Augmenting Motor Recovery Following Stroke. Congress of Neurological Surgeons, *Los Angeles, CA USA*, Oct 2025, *Poster presentation by* Khazanchi R

Hirsch ZM, **Tillman M**, Liu JM, Molino J, Zaferiou AM. Older Adult Frontal Plane Angular Momentum and Lateral Distance During 90 Degree Turns While Walking. Society for Biomechanics, Bioengineering, and Biotransport Conference, *Santa Ana Pueblo, NM USA*, Jun 2025, *Poster presentation by* Hirsch ZM

Spinal motor evoked responses with transcutaneous spinal cord stimulation: Influence of carrier frequency, waveform, electrode location and configuration, and demographic characteristics on the resting motor thresholds and tolerance of stroke. North American Neuromodulation Society, *Arlington, VA USA*, Jun 2025, *Poster presentation*

2023 **Tillman M**, Zaferiou AM. Momenta control during each phase of gait reveals a common strategy between straight-line gait and 90-degree turns. Society for Neuroscience, *Washington D.C. USA*, Nov 2023, *Poster presentation by* **Tillman M**

Hirsch ZM, **Tillman M**, Zaferiou AM. Frontal Plane Balance Patterns of Older Adults During Pre-planned and Late-cued Turns. American Society of Biomechanics, *Knoxville TN USA*, Aug 2023, *Poster presentation by* Hirsch ZM

Tillman M, Zaferiou AM. Gait Phase-Specific Linear and Angular Momentum Generation During 90-Degree Left Turns in Healthy Older Adults. International Society of Posture and Gait Research World Congress, *Virtual Brisbane AU*, Jul 2023, *Poster presentation by* **Tillman M**

Iyer V, **Tillman M**, Zaferiou AM. Evaluating the Understandability of Real-Time Sonified Biofeedback Prototypes for Balance Training. Summer Biomechanics Bioengineering and Biotransport Conference, *Vail CO USA*, Jun 2023, *Poster presentation by* Iyer V

2022 **Tillman M**, Zaferiou AM. Generation of Transverse Plane Linear and Angular Momenta During 90 Degree Turns. American Society of Biomechanics & North American Congress on Biomechanics Joint Conference, *Ottawa ON CA*, Aug 2022, *Podium presentation by* **Tillman M**

Tillman M, Zaferiou AM. Lateral Distance as a Measure of Balance During 90 Degree Pre-Planned and Late-Cued Turns. Summer Biomechanics Bioengineering and Biotransport Conference, *Cambridge MA USA*, Jun 2022, *Student-competition finalist; Podium presentation by* Dr. Zaferiou

Tillman M, Zaferiou AM. Gait Phase-Specific Generation of Transverse Plane Momenta during Pre-planned and Late-cued Turns. Northeast Bioengineering Conference, *NYC NY USA*, Apr 2022, *Podium presentation by* Dr. Zaferiou

2021 **Tillman M**, Zaferiou AM. User-Centered Design of Sonified Balance Biofeedback. International Conference of the IEEE Engineering in Medicine and Biology Society, *Virtual*, Oct 2021, *Podium presentation by* Zaferiou A

Tillman M, Zaferiou AM. Frontal-Plane Whole-Body Angular Momentum During Pre-Planned and Late-Cued Turns. American Society of Biomechanics, *Virtual*, Aug 2021, *Podium presentation by* **Tillman M**

2020 **Tillman M**, Dahl L, Knowlton CB, Zaferiou AM. Early Testing of a Real-Time Balance Sonification System In Single Leg Stance. American Society of Biomechanics, *Virtual*, Aug 2020, *Oral presentation by* **Tillman M**

2019 **Tillman M**, Ambike S. The Stability of the Current Motor State is Influenced by Expected Movement: Do Cognitive Events During the Inter-Stimulus Interval of Choice Reaction-Time Tasks Have a Motor Counterpart? Society for Neuroscience, *Chicago IL USA*, Oct 2019, *Poster presentation by* **Tillman M**

Munoz-Ruiz M, **Tillman M**, Ambike S. Interactions between fingers during rapid force pulse production. American Society of Biomechanics & International Society of Biomechanics Joint Conference, *Calgary AB Canada*, Aug 2019, *Poster presentation by* Munoz-Ruiz M

Tillman M, Ambike S. Effects of Past and Future Motor Actions on Present Multifinger Pressing Behavior. American Society of Biomechanics & International Society of Biomechanics Joint Conference, *Calgary AB Canada*, Aug 2019, *Poster presentation by* **Tillman M**

Tillman M, Ambike S. Effects of Past and Future Motor Actions on Present Multifinger Pressing Behavior. American Society of Biomechanics Regional Conference, *Dayton OH USA*, Feb 2019, *Podium presentation by* **Tillman M**

2018 **Tillman M**, Ambike S. Mechanisms of Preparation for Task Switching in a Finger Pressing Task. American Society of Biomechanics, *Rochester MN USA*, Aug 2018, *Poster presentation by* **Tillman M**

Tillman M, Ambike S. Reduction in stability of manual behavior in uncertain conditions. Purdue Undergraduate Research Symposium, *West Lafayette IN USA*, Apr 2018, *Poster presentation by* **Tillman M**

2017 **Tillman M**, Ambike S. Reduction in stability of manual behavior in uncertain conditions. American Society of Biomechanics, *Boulder CO USA*, Aug 2017, *Poster presentation by* Dr. Ambike

Tillman M, Ambike S. Uncertain motor plans lead to reduced stability of the current state in young but not in older adults. Progress in Motor Control, *Miami FL USA*, Jul 2017, *Poster presentation by* Dr. Ambike

Tillman M, Ambike S. Uncertain motor plans lower stability of current prehensile behaviour. Purdue Institute of Integrative Neuroscience Retreat, *St. Joseph MI USA*, May 2017, *Poster presentation by* Dr. Ambike

Tillman M, Ambike S. Uncertain motor plans lower stability of current prehensile behaviour. Purdue Undergraduate Research Symposium, *West Lafayette IN USA*, Apr 2017, *Poster presentation by* **Tillman M**

Tillman M, Ambike S. Uncertain motor plans lower stability of current prehensile behaviour. American Society of Biomechanics Regional Conference, *Grand Rapids MI USA*, Feb 2017, *Podium presentation by* **Tillman M**

Additional Presentations

2022 **Tillman M**. Problems and Ideas Relating to Sharing Code in Research. American Society of Biomechanics Student Seminar Series, *Virtual*, Dec 2022, *Oral presentation by* **Tillman M**

Tillman M. sciOS a "science operating system": Developing A Software Platform to Facilitate Scientific Analysis. Department Seminar, *Hoboken NJ USA*, Feb 2022, *Oral presentation by* **Tillman M**

Tillman M. The Role of Each Leg In Turn Performance. Department Seminar, *Hoboken NJ USA*, Feb 2022, *Oral presentation by* **Tillman M**

2021 **Tillman M**. The Biomechanics of Frontal Plane Balance. Department Seminar, *Hoboken NJ USA*, Sep 2021, *Oral presentation by* **Tillman M**

Leadership

2021 **Secretary**, *Graduate Biomedical Engineering Society*, 2021-2022

- Founded a student-led weekly seminar series for PhD students and postdocs to share their research and network with other students and faculty in the department.

- Organized multiple career preparedness events per semester.

2020 **Summer Undergraduate Research Program**, *Stevens Institute of Technology*, 2020-2024

- Led undergraduate students through a 10-week summer research program

- Led meetings, organized and developed student projects, and provided mentorship

2019 **Graduate Student Training**, *Purdue University, Biomechanics Lab*, 2019

- Instructed graduate students on data collection protocols.

2018 **Vice-President**, *HK GSO, Purdue University*, 2018-2019

- Vice President of Purdue University's Department of Health & Kinesiology Graduate Student Organization (HK GSO).

- Organized departmental fundraisers and professional development workshops.

Service

2024 **National Biomechanics Day**, Apr 2024

- Taught middle and high school students foundational biomechanics concepts using engaging activities and laboratory equipment (force plates, motion capture, IMU's, etc.)

2023 **Computer Science Undergraduate Student Senior Design Client**, 2023-2024

- Provided a computer science undergraduate senior design team with their senior design idea: developing a 3D biomechanical data visualizer.

- During weekly meetings I guided their design process by informing the team of the needs and use cases of biomechanics researchers regarding data visualization, and gave feedback on their implementation.

National Biomechanics Day, Softball, May 2023

- Led undergraduate students in preparing activities to teach high school softball players about biomechanics concepts related to softball.

2022 **American Society of Biomechanics Student Committee Member**, 2022-2024

- Secured a sponsor to fund the 2023 and 2024 ASB student night out event

- Participated in regular committee meetings, and student committee decisions regarding activities during the year and the conference itself

National Biomechanics Day, Dance, May 2022

- Prepared and led activities to teach biomechanics concepts to high school students at Frank Sinatra School of the Arts in Queens, NY

2021 **Recruitment of Department Chair, Faculty, and Graduate Students**, 2021-2023

- Participated in faculty and department chair interviews by attending seminars and prospective faculty lunches (approximately 5 per year)

National Biomechanics Day, Dance, Apr 2021

- Hosted a live dance performance using the real-time auditory biofeedback system to educate high school students on biomechanics

Awards

2023 **Doctoral Excellence Fellowship**, Stevens Institute of Technology, May 2023, *Full tuition and stipend awarded for one academic year based on excellence in activities throughout Ph.D. program*

2022 **Robert Crooks Stanley Fellowship**, Stevens Institute of Technology, Jul 2022, *Full tuition and stipend awarded for one academic year based on excellence in research activities in accordance with the values set forth by the Stanley Fellowship committee*

2019 **Outstanding Masters Student in the College of Health and Human Sciences**, Purdue University College of Health and Human Sciences, Apr 2019, *$500 awarded for outstanding achievements in original research*

Outstanding Masters Student in the Department of Health and Kinesiology, Purdue University Dept. of Health and Kinesiology, Mar 2019, *$200 awarded for outstanding achievements in original research*

2018 **Outstanding Senior in Movement and Sports Sciences**, Purdue Dept. of Health and Kinesiology, Apr 2018, *Awarded for outstanding engagement in original undergraduate research under the guidance of a Purdue faculty mentor*

Office of Undergraduate Research Scholarship, Purdue University Office of Undergraduate Research, Spr. 2018, *$1,000 awarded for engagement in original undergraduate research under the guidance of a Purdue faculty mentor*

2017 **Outstanding Undergraduate Research Assistant**, Purdue College of Health and Human Sciences, Apr 2017, *Awarded to recognize exceptional performance in the course of assisting a faculty member's research*

Ray Anne Shrader Undergraduate Travel Scholarship, Purdue Dept. of Health and Kinesiology, Jan 2017, *$500 awarded for undergraduate academic travel. Covered the cost of attending the Midwest American Society of Biomechanics (ASB) regional conference in Grand Rapids, MI*

Certificates

2022 **Teaching at the College Level**, Stevens Institute of Technology, Apr 2022, *8 week course covering the fundamentals of academic teaching and course design*

Professional Affiliations

2023 International Society of Posture and Gait Research, *Student Member*, 2023-Present

2021 Graduate Biomedical Engineering Society, *Member*, 2021-2022

2019 Society for Neuroscience, *Graduate Student Member*, 2019, 2023

2017 American Society of Biomechanics, *Student Member*, 2017-Present